Quantum Mechanics II — Exercise Sheet 5

Spectral readings and conditioning

Variant: Solutions

Exercise 1 (From momentum to energy density). A normalized line state has momentum density $(2b)^{-1}\mathbf{1}_{[-b,b]}$, where $b > 0$. (i) Find the cumulative distribution and density of its free energy $H_0 = P^2/(2m)$. (ii) Compute the median energy and the mean energy from this density.

Solution. For $0 \le E \le b^2/(2m)$ the event is $|p| \le \sqrt{2mE}$. The cumulative probability and its derivative are

$$F(E) = \frac{\sqrt{2mE}}{b}, \qquad w(E) = \frac{\sqrt{2m}}{2b\sqrt{E}}.$$

The cumulative distribution is zero below zero and one above $b^2/(2m)$; the density vanishes outside this interval. Setting $F(E) = 1/2$ and integrating $Ew(E)$ give

$$E_{\mathrm{med}} = \frac{b^2}{8m}, \qquad \langle H_0 \rangle = \frac{b^2}{6m}.$$

The integrable density singularity at zero is not an atom.

Exercise 2 (A degenerate reading retains a phase). Let $f \in L^2(\mathbb{R})$ be normalized and supported in $(a, 2a)$ with $a > 0$. Set $g(x) = f(-x)$ and $\psi = (f + e^{i\phi} g)/\sqrt{2}$. Compare a reading of Q^2 in $(a^2, 4a^2)$ with a reading of the sign of Q whose result is forgotten. Give the resulting states and the expectation of parity $(J\chi)(x) = \chi(-x)$ in each.

Solution. The supports of f and g are disjoint, so they are orthonormal. The Q^2 event has probability one and leaves the pure vector ψ unchanged. The forgotten sign reading leaves the density operator

$$\rho = \frac{1}{2}|f\rangle\langle f| + \frac{1}{2}|g\rangle\langle g|.$$

Since $Jf = g$ and $Jg = f$, the two parity means are

$$\langle \psi, J\psi \rangle = \cos\phi, \qquad \mathrm{Tr}(\rho J) = 0.$$

A degenerate spectral projection need not destroy the relative phase inside its range.

Exercise 3 (A joint reading that is not independent). A normalized state has position density $1/(2a)$ on $(-a, a)$ and zero elsewhere. The commuting observables are Q and Q^2. (i) Find $\mathrm{Prob}(Q > 0, Q^2 < a^2/4)$ and compare it with the product of the marginal probabilities. (ii) Then replace $Q > 0$ by $Q > a/2$ and decide whether the two observables are independent in this state.

Solution. Joint spectral projections multiply the indicator functions of the two conditions. For the first pair the intersection is $(0, a/2)$, so

$$\mathrm{Prob}(Q > 0, Q^2 < a^2/4) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}.$$

For the second pair the intersection is empty, whereas

$$\mathrm{Prob}(Q > a/2) = \frac{1}{4}, \qquad \mathrm{Prob}(Q^2 < a^2/4) = \frac{1}{2}.$$

The joint probability is zero and the product is $1/8$. Equality for one pair of events does not establish independence.

Exercise 4 (A finite first moment without a second). Take position density $w(x) = (1+|x|)^{-3}$ and the normalized vector $\psi = \sqrt{w}$. Check normalization, compute the mean position, and decide whether $\psi \in D(Q)$. Distinguish an absolutely convergent mean from a symmetric principal value.

Solution. The integral of the density is $2\int_0^\infty (1+x)^{-3}\,\mathrm{d}x = 1$. Its absolute first moment is

$$\int_{\mathbb{R}} |x| w(x)\,\mathrm{d}x = 2\int_0^\infty \frac{x}{(1+x)^3}\,\mathrm{d}x = 1.$$

The mean exists as an absolutely convergent integral and is zero by parity. The second-moment integrand has order $1/x$ at infinity, so

$$\int_{\mathbb{R}} x^2 w(x)\,\mathrm{d}x = \infty, \qquad \psi \notin D(Q).$$

No principal-value cancellation is used to define the mean.